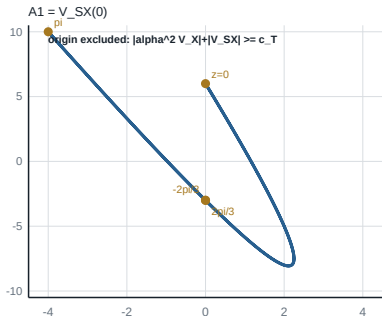
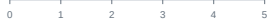


A exact cell geometry and no-go ratios



centered ratios at $y=\pi$



H5/4 4.112

H5+H7 1.570

exact-H3 1.135

tail after H7 2.705

time-only scalar gauges already removed

B compact versus escaping transfer

Taylor lane

compact chart: $\alpha R \rightarrow 0$

$V_{\text{exact}} = H3 + \text{local weighted remainder}$

bounded-multiplier absorption

only $R = o(\alpha^{(-2/5)})$

Escaping / torus lane

$R = y/\alpha$: every Taylor term is leading

centered H5/4 ratio $\rightarrow 5 y^2 / 12$

global termwise absorption: FALSE

infinite trigonometric cancellation retained

exact compact-escaping cell theorem

unit cells $\rightarrow$ full $H^{\sim 1}$ direct sum

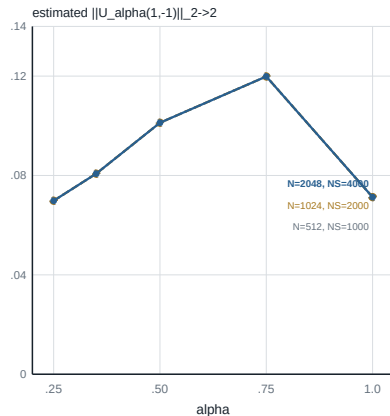
$\rightarrow$ expanding torus graph estimate

exact periodic scalar-row block

strict contraction: CLOSED

exact periodic scalar-row block contraction: CLOSED

C forward-adjoint norm diagnostic



diagnostic ledger

max relative-to-fine: $1.19e-04$

max adjoint defect: $3.59e-13$

max power residual: $1.89e-13$

Fourier Strang + fixed power iteration on U^*U

NUMERICAL DIAGNOSTIC ONLY — NOT PROOF

outer concatenation / nonlinear / Clay: OPEN

diagnostic does not evaluate C_T